Pan and the Encke Gap Edge Waves: a Farey–Ford Reading of a Resonance Ladder

An application example of the Hyperbolic Tropical Wavefront–Residual (HTWR) framework

Mingli Yuan*
with computational assistance from DeepSeek Harness

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Abstract

The scalloped edges of Saturn's Encke gap are the gravitational signature of the shepherd moon Pan. We give a two-layer account. The *arithmetic layer* reads Pan's Lindblad resonance ladder $x = m/(m \pm 1)$ as the Farey neighbourhood of corotation ($x = 1$), realized as Ford circles piling up on the hyperbolic upper half-plane — a concrete physical instance of the rational-approximation structure at the core of the HTWR framework. The *dynamical layer* then resolves the observed wave: it is neither a propagating density wave nor a (cold or pressure) edge mode, but Pan's *static tidal distortion* (a forced epicycle), with radial amplitude $GM_P/(\kappa^2 d^2) \approx 0.79\,\mathrm{km}$ and azimuthal scale $\approx 250\,\mathrm{km}$, matching the values extracted from a Cassini image ($\approx 0.65\,\mathrm{km}$, $\approx 300\,\mathrm{km}$). The two layers are complementary: the arithmetic skeleton locates the gap (resonance overlap), while the tidal distortion shapes its edge.

Keywords: planetary rings; Encke gap; Pan; Lindblad resonances; Farey sequence; Ford circles; epicyclic theory; Chirikov resonance overlap.

Status. §2–§3 (the Farey ladder and the gap-opening criterion) are the HTWR reading, recorded as *structural proposals* plus elementary celestial mechanics. §4–§6 are standard linear celestial mechanics, labelled PROVED WITH STATED HYPOTHESES or COMPUTATIONALLY VERIFIED EXAMPLE as appropriate. This note claims no new theorem of AEG; it exhibits a physical realization of the HTWR rational-approximation structure.

1 Introduction

Pan (mass $M_P \approx 4.95 \times 10^{15}\,\mathrm{kg}$, semi-major axis $a_P = 133{,}584\,\mathrm{km}$) orbits inside the Encke gap, a 325-km-wide division in Saturn's A ring. Its gravity excites a scalloped ("wavy") pattern on both gap edges, observed by Voyager and Cassini and ubiquitous over 360° of longitude [4]. The aim of this note is twofold: (i) to exhibit the *arithmetic structure* of Pan's resonances — a Farey ladder accumulating at corotation — as a concrete instance of the HTWR framework's rational-approximation geometry; and (ii) to resolve the *dynamical mechanism* of the observed edge wave by elimination, ending at Pan's static tidal distortion. The two parts are complementary, not the same

*Corresponding author and sole holder of scientific responsibility. Computation, drafting, and writing were assisted by DeepSeek Harness, an artificial-intelligence research agent; that assistance is credited here but is not human authorship, since the agent is not a natural person and cannot sign a copyright transfer.

phenomenon: the resonance ladder explains *why the gap is there*, the tidal distortion explains *why its edge is scalloped*.

2 The resonance ladder is a Farey sequence (the arithmetic layer)

In the linear epicyclic theory, the m-th gravitational component of Pan (pattern speed Ω_P) excites a ring particle at the Lindblad resonance

$$m(\Omega - \Omega_P) = \pm\kappa, \qquad \Omega = \kappa = \Omega_P\,(r/a_P)^{-3/2}. \tag{1}$$

Writing $x := \Omega/\Omega_P$, this is $m(x-1) = \pm x$, i.e.

$$\text{inner } (r < a_P):\ x = \frac{m}{m-1}, \qquad \text{outer } (r > a_P):\ x = \frac{m}{m+1}, \qquad \frac{r}{a_P} = x^{-2/3}. \tag{2}$$

Hence the resonance *frequencies* are exactly the Farey neighbours of corotation, $x = m/(m \pm 1) \to 1$ as $m \to \infty$.

m	outer $x = m/(m+1)$	r/a_P (Arb-certified)	Ford radius $1/(2q^2)$
1	1/2	1.58740	1.25×10^{-1}
10	10/11	1.06560	4.13×10^{-3}
100	100/101	1.00666	4.90×10^{-5}
551	551/552	1.00121	1.64×10^{-6}

Table 1: The outer Lindblad resonance ladder of Pan.

At $m = 551$ the outer resonance sits at $r = a_P + 161.6\,\text{km}$, i.e. at the Encke outer edge ($+162\,\text{km}$). This is the HTWR reading in one line: the resonance ladder is a sequence of rational points $p/q \in \mathbb{P}^1(\mathbb{Q})$, each carrying a Ford circle of radius $1/(2q^2)$ tangent to the real axis at p/q, piling up at the corotation cusp $x = 1$; the m-th resonance has "information height" $I = \ln\max(p,q) = \ln(m+1)$ (the logarithm of the azimuthal wavenumber).

Status. The ladder identity (2) is elementary (PROVED). The Ford-circle / hyperbolic reading is a STRUCTURAL PROPOSAL, not a new theorem.

3 Gap opening: the Chirikov resonance-overlap criterion

The gap edge is where the high-m resonances *overlap*. The relevant scales are

$$R_H = a_P(\mu/3)^{1/3} \approx 19.0\ \text{km} \quad (\mu = M_P/M_S = 8.7 \times 10^{-12}), \qquad \delta r \approx a_P\sqrt{\mu} \approx 0.39\ \text{km}, \tag{3}$$

and the resonance spacing $dR_m = 2a_P/(3m^2) = 0.293\,\text{km}$ at $m = 551$. Overlap (Chirikov) requires $\delta r \gtrsim dR_m$, i.e. $m \gtrsim \sqrt{2/(3\sqrt{\mu})} \approx 475$. The observed edge ($m \approx 551$) agrees with this to the $\mathcal{O}(1)$ coefficient of the resonance width. Since gap$/R_H \approx 8.5$, the gap is opened by the *resonance forest* (the dense high-m ladder), not by direct Hill-sphere clearing.

4 The density-wave dispersion (the free wave, eliminated)

The tight-winding (WKB) dispersion for a self-gravitating, pressure-supported disk is

$$(\omega - m\Omega)^2 = \kappa^2 - 2\pi G\Sigma|k| + c^2k^2, \tag{4}$$

a closed-form quadratic in the radial wavenumber k. At resonance the radial wavelength is $\lambda_0 = c^2/(G\Sigma) \approx 0.75\,\mathrm{km}$ ($\Sigma = 500\,\mathrm{kg/m^2}$, $c = 0.5\,\mathrm{cm/s}$), *decreasing* away from resonance (the "wrapped-up" spiral). This matches the observed "wavelength varies smoothly" signature qualitatively, but the free wave lives at $m < 551$ (resonance in the ring); the measured edge wave is $m \approx 2800$, whose resonance lies *inside* the cleared gap. Hence the observed edge wave is not the free density wave.

5 Edge-mode elimination (cold and pressure)

We test the Papaloizou–Lin edge mode in the cold WKB limit and with pressure.

Cold WKB ($\sigma^2 = \kappa^2 - 2\pi G\Sigma|k|$, reflecting edge, Airy quantization $\int |k|\,\mathrm{d}r = (n + 3/4)\pi$): the turning point $r_L = a_P((m+1)/m)^{2/3}$ lies outside the edge only for $m < 551$; for $m \geq 551$ the resonance is inside the gap and the edge is far off-resonance (evanescent decay length $\approx 0.5\,\mathrm{m}$).

Pressure ($Q = c\kappa/(\pi G\Sigma) \approx 6$): the existence condition is geometric ($m < 548$), independent of c; at $m = 2800$ the pressure only yields a short wave of radial wavelength $\approx 47\,\mathrm{m}$.

So the observed edge wave is neither a density wave nor an edge mode (cold or pressure).

6 The static tidal distortion (the resolution)

The edge lies $d \approx 162\,\mathrm{km} \approx 8.5\,R_H$ from Pan — the linear tidal regime ($d \gg R_H$). Pan's gravity pulls the edge radially, balanced by the epicyclic restoring force $\kappa^2\delta r$:

$$\kappa^2\,\delta r = -\frac{GM_P\,d}{(d^2 + y^2)^{3/2}}, \tag{5}$$

where y is the azimuthal offset from Pan. This is a *forced epicycle* with eccentricity

$$e_f = \frac{M_P}{M_S}\Big(\frac{a_P}{d}\Big)^2 \approx 5.9 \times 10^{-6}, \tag{6}$$

radial amplitude $A = a_P\,e_f = GM_P/(\kappa^2 d^2)$, and azimuthal profile $\delta r(y) = \delta r_{\max}\,(1 + (y/d)^2)^{-3/2}$ (a localized bump of width d).

quantity	computed	observed (PIA08257)
radial amplitude A	$0.785\,\mathrm{km}$	$\approx 0.65\,\mathrm{km}$
azimuthal FWHM	$249\,\mathrm{km}$	$\approx 300\,\mathrm{km}$

Table 2: Static tidal distortion versus the image extraction.

Both agree to $\sim 20\%$, inside the Σ/c/geometry uncertainty.

7 Observation: extraction from a Cassini image

We extracted the edge-wave parameters directly from NASA's public-domain image PIA08257 ("Encke Waves", 1 km/px original, stretched 4× in the radial direction), detecting the gap edges and fitting the strongest sinusoid: azimuthal wavelength $\lambda_\theta \approx 260$–$320\,\mathrm{km}$ and radial amplitude $A \approx 0.65\,\mathrm{km}$, hence $m \approx 2800$ (from $m = 2\pi r/\lambda_\theta$). Reproducible in `aeglib`/numpy scripts (`008-encke-edge-wave-image.py`).

8 Discussion: what the HTWR framework does and does not contribute

The HTWR framework contributes the *arithmetic skeleton*: the resonance ladder is a Farey sequence realized as Ford circles on $\mathbb{H}^2$, with the corotation point as the cusp. This is a *physical realization* of the rational-approximation structure that the HTWR axioms (state space $\mathbb{P}^1(\mathbb{Q})$, logarithmic-Weil-height calibration) abstract — not a new theorem, and not the mechanism of the edge wave itself.

The edge-wave *mechanism* is standard linear celestial mechanics (forced epicycle), where the HTWR framework has no role. The two layers answer different questions: *where* the gap is (resonance overlap, the Farey pile-up), and *what shape* its edge has (static tidal distortion).

Status. Honest boundary. The secondary signatures — "wavelength varies smoothly", "amplitude grows and decays", "undamped over $\gtrsim 10$ months" — are not captured by the static force balance; they require the full dynamics (collisional damping and multi-m superposition). These remain open (`governance/08-open-questions.md`, OQ-104).

9 Conclusion

The Encke gap's scalloped edges are a two-layer phenomenon: an *arithmetic skeleton* (Pan's Lindblad resonances form the Farey ladder $x = m/(m \pm 1)$, accumulating at corotation as Ford circles — a concrete instance of the HTWR rational-approximation structure), and a *dynamical layer* (the observed edge wave is Pan's static tidal distortion, a forced epicycle of amplitude $GM_P/(\kappa^2 d^2) \approx 0.79\,\mathrm{km}$, matching the observed $\approx 0.65\,\mathrm{km}$). The framework's value here is exemplary: it turns a resonance catalogue into a geometric object on $\mathbb{P}^1(\mathbb{Q})$, while the physics that shapes the edge remains classical.

Acknowledgements and provenance

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A Mathematical status register

ID	status
R1	resonance ladder $x = m/(m \pm 1)$, radius $r/a_P = x^{-2/3}$ — PROVED WITH STATED HYPOTHESES
R2	Ford-circle / hyperbolic reading (HTWR) — STRUCTURAL PROPOSAL
R3	Chirikov overlap $\to$ gap edge $m \approx 551$ — STANDARD CONSEQUENCE + COMPUTATIONALLY VERIFIED
R4	density-wave dispersion, $\lambda_0 = c^2/(G\Sigma)$ — PROVED WITH STATED HYPOTHESES
R5	edge mode absent for $m \geq 551$ (cold & pressure) — PROVED WITH STATED HYPOTHESES
R6	static tidal distortion $A = GM_P/(\kappa^2 d^2)$ — PROVED WITH STATED HYPOTHESES
R7	image extraction $\lambda_\theta \approx 300\,\mathrm{km}$, $A \approx 0.65\,\mathrm{km}$ — COMPUTATIONALLY VERIFIED EXAMPLE

References

[1] Jeffrey N. Cuzzi and Jeffrey D. Scargle. Wavy edges suggest moonlet in Encke's gap. *The Astrophysical Journal*, 292:276–290, 1985.

[2] P. Goldreich and S. Tremaine. The excitation of density waves at the Lindblad and corotation resonances by an external potential. *The Astrophysical Journal*, 233:857–871, 1979.

[3] J. C. B. Papaloizou and D. N. C. Lin. On the pulsational modes of a sharp-edged ring. *The Astrophysical Journal*, 331:838–860, 1988.

[4] M. S. Tiscareno, J. A. Burns, M. M. Hedman, J. N. Spitale, C. C. Porco, and C. D. Murray. Wavy edges and other disturbances in Saturn's Encke and Keeler gaps. *Bulletin of the American Astronomical Society*, 37:64.02, 2005.